

Field Hockey

● ● ● ● ● ● ● ●

Dave Hamilton

Performance Demands of Field Hockey

First played in 19th century Great Britain, the sport of field hockey is now enjoyed in both indoor and outdoor settings. The latter format is far more popular and became a permanent Olympic staple in 1928. Field hockey is similar to many field-based team sports in that it requires a blend of technical, tactical, and physical variables to perform well and be successful. This section will introduce the sport of field hockey with focused discussion on performance variables at the competitive level.

Sport Description and Factors of Winning

Field hockey is a team sport played by both men and women. The game has indoor and outdoor formats, but it is the traditional outdoor format that draws the biggest audiences with the two largest tournament-based competitions being the Olympic Games and World Cup. The outdoor teams are made up of 11 players, normally 10 field players and 1 goalkeeper. The field dimensions are not dissimilar to those of other field sport cousins (soccer, rugby, lacrosse), with the dimensions being 91 × 55 m (100 × 60 yd) (figure 25.1). The modern game is played on a synthetic water-based pitch with arti-

cial grass, which serves to reduce friction between all stick-to-turf and ball-to-turf activities. This water-based feature has facilitated a faster game and increased the finer motor skill demands of the athletes as it pertains to stick-and-ball control through hand dexterity and hand-eye coordination.

The 11 players on a team can typically be classified into four categories that include forwards, midfielders, backs, and goalkeepers (figure 25.1). While each position requires slightly different skillsets, all players are free to roam the entire field, with no offside rule ever called during play. Scoring can occur only when players strike the ball within the opponent's shooting circle, called the D. Rolling substitutions with the 11 on-field players and the 7 benched players are a defining component of the game, as there is no limit to the number of substitution rotations that can be completed. This creates the opportunity to maintain intermittent levels of high intensity throughout the game, much like a high-intensity interval training (HIIT) format. In 2014, the International Hockey Federation game format was changed from 2 × 35 min halves with a 10 min half-time, into 4 × 15 min quarters, with 2 min rest periods and one 10 min halftime. In simple terms, today's format consists of a 60 min game that includes four 15 min quarters.

Field hockey is a highly demanding intermittent effort sport requiring myriad high-intensity actions.

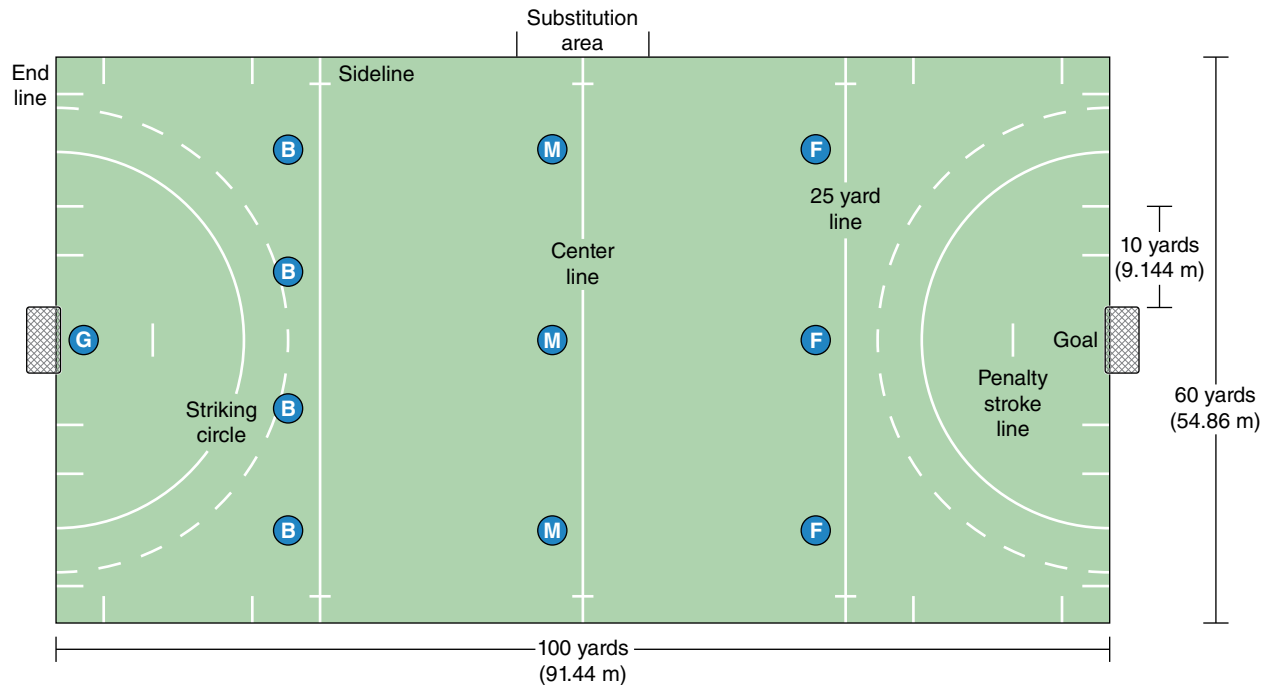


Figure 25.1 Field hockey field diagram, including dimensions and positions. F: forwards; M: midfielders; B: backs; G: goalkeepers.

Global positioning system (GPS) analysis has shown that outfield players can cover up to 5 to 9 km per game. It is therefore appropriate to view field hockey as an aerobically demanding sport, with frequent although brief superimposed anaerobic efforts. High-intensity actions contribute to a significant percentage of match activities (12% to 26%), and it is the successful completion of these high-intensity actions that can often determine the outcome of a field hockey match (6). It is clear that hockey by nature requires a high level of aerobic fitness, a quality that when well developed will facilitate an athlete's ability to cover greater distances during a match and recover effectively between repeated high-intensity efforts.

Relative Contribution of Physical Performance

The demands of field hockey match play and intensity are typically higher than those described in other field-based sports, an outcome inherently created through the rulings of unlimited rolling substitutions, no offside, and the self-pass rule from free hits (5). The purpose of such rule modifications was to improve the spectacle of the game for the audience by reducing stoppages and increasing the relative time the ball was in play. As a result, the modern game has created a greater reliance on not only

technical and tactical capabilities but physical qualities as well that underpin an athlete's ability to repeat these high-intensity activities throughout a game (figure 25.2). Having athletes who can endure and tolerate repeated bouts of low-to-ground actions, high-speed running, rapid changes of direction, and forceful accelerations and decelerations is paramount for success. Indeed, the ability to display these speed, strength, and power qualities throughout the game, while mitigating excessive fatigue, has been shown to play a crucial role alongside endurance as key determinants of field hockey success (7, 10). Like many field or court-based sports, successful outcomes and winning can heavily depend on the ability to sustain these actions consistently throughout a game and/or season.

It's clear that an athlete who is physically strong and aerobically fit offers the coach a springboard from which to develop an impactful player, but the demands of international hockey are very specific. With club and league hockey systems, 1 or 2 games are played each week over a 20 to 30 wk season. Group games, back-to-back games, knockout stages, shoot-outs, and the need to perform for 8 games in 12 d during Olympic and World Championship tournaments are the keys to success at the international level. The physical resilience and conditioning levels required for these specific competition stresses



Figure 25.2 The position of the field hockey player on the three axes illustrates the relative importance of the three main physical capacities of importance for elite participation in field hockey, acknowledging this varies slightly between positions. Outside of critical psychological aspects and team chemistry components, which are difficult to quantify, the pie chart shows the general relative importance of skills (40%), tactical awareness (25%), and physical capacities (35%) for field hockey success.

Adapted from G.A. Nader, "Concurrent Strength and Endurance Training: From Molecules to Man," *Medicine & Science in Sports & Exercise* 38, no. 11 (2006): 1965-1970.

undoubtedly bring unique challenges to the way we prepare athletes. With this in mind, the goal of any physical preparation program should be to develop an athlete who can tolerate high workloads and is robust and durable enough to endure the rigors of tournament field hockey. The focus then shifts to developing a training approach that systematically improves tolerance levels, so that athletes can endure the high training volumes required to succeed in competitions such as the Olympic Games.

Repeating high-intensity actions and having the ability to express high-force outputs is crucial and underpins the opportunity for improved success. We know that athletes who possess high relative strength are capable of expressing higher forces and superior levels of speed and power. Field hockey is a sport in which success depends on players winning individual battles. The ability to accelerate effectively and perform rapid unilateral movements in and out of low-to-ground positions are key characteristics that may help shape successful outcomes against counterparts.

Benchmarking

Benchmarking is the process of identifying the key physical performance characteristic needed to be successful in your sport, as well as offering insight into the type of athlete you have at your disposal.

It additionally provides your athlete with clear guidelines on her physical development as it relates to her sport. It is imperative that the tests chosen for benchmarking are valid, standardized, and reflect the physical performance characteristics deemed conducive for success in field hockey, thus allowing for the amalgamation of normative data, future cross comparisons, and the highlighting of key performance indicators. An example of my benchmarking is shown in table 25.1.

Figure 25.3 shows the benchmarking of physical qualities identified as important in elite female field hockey players. These standards are based on normative data collected from a number of elite female athlete populations across a variety of field and court-based sports (rugby, soccer, netball, tennis, and lacrosse). For more unique assessments with limited data sets, the use of the standard deviation (SD) away from the mean can be a useful process to help clarify benchmarking categories. The physical tests outlined in table 25.1 and figure 25.3 are designed to highlight an athlete's current level of development as it relates to elite-level counterparts.

Benchmarking also gives the athlete clear objective goals for which to strive. These chosen tests allow for a simplified overview of all the physical qualities deemed essential for athletic prowess in

Table 25.1 Benchmarking Tests and the Variables Assessed

Physical quality	Test	Variable assessed
Gym-based strength	Back squat, bench press, wide chins, and jump squat profiling	2 to 5 RMs
Strength	Isometric mid-thigh pull	Rate of force development Absolute and relative strength
Power	Countermovement jump squat jump	Dynamic force qualities Eccentric utilization ratio
Speed	10, 20, 40 m	Acceleration, peak velocity
Anaerobic endurance	6 × 40 m (20 m shuttle)	Fatigability
Aerobic power	30-15 intermittent fitness test (30-15 IFT)	High-intensity intermittent performance VO ₂ max (predicted) Max heart rate
Movement quality	Movement assessment	Flexibility control through ROM Kinetic chain efficiency and neuromuscular system
Anaerobic speed reserve	30-15 intermittent fitness test (30-15 IFT)	Anaerobic speed reserve (peak velocity attained in 30-15 IFT)

	Speed				Strength		Conditioning		
	5 m	10 m	40 m	ISO pull	Back sq	Bench press	Wide pull-ups	Mean RSA	30-15 _{IFT}
Elite and 2012	0.99	1.75	5.4	3.75	2	1.05	12	7.2	21
Green	1.04	1.8	5.65	3	1.75	0.9	9	7.45	20
Amber	1.1	1.87	5.85	2.4	1.5	0.75	6	7.7	19.5
Red	1.15	1.93	6	2	1.25	0.6	3	7.9	19

Figure 25.3 Athlete profile traffic light, showing the benchmarking of physical qualities identified as important for elite female field hockey players. Speed 5, 10, 40 m=time (s); strength back squat and bench=relative strength (× body mass); pull-up=repetitions; repeated sprint ability (RSA)=average time across 6 sprints; 30-15 IFT=level reached.

field hockey. This traffic light system (figure 25.3) has been put together so that athletes understand where they are and how much they need to develop when competing at the world-class elite level. Figure 25.3 shows the physical level and expectation of field hockey athletes leading into an Olympic campaign. Typically these tests are performed pre-season, mid-season, and 6 wk prior to the first major international competition, allowing sufficient time for any prescriptive changes to address specific physical qualities.

Field Tests and Order Performed

DAY 1

Speed: 10, 20, 30, 40 m; acceleration qualities and peak velocity established

Aerobic power: 30-15 IFT (2); establish an athlete's aerobic and anaerobic power, recovery, and change of direction (COD) abilities

DAY 2

Anaerobic power and agility: Hamilton 6 × 40 m RSA (see drill protocol and figure 25.4); highlight anaerobic qualities and fatigability of the athletes

Strength test: power (countermovement jump [CMJ] and squat jump [SJ]); relative power and dynamic force qualities analyzed; eccentric utilization ratio (EUR) can also be established (8)

The schematic shown in figure 25.2 offers a perceptual guide as to where field hockey demands sit in regard to their physiological demands and where

Hamilton RSA Protocol for Field Hockey Players

The Hamilton RSA protocol for field hockey players consists of a 20-20 shuttle with an active 20 m recovery. It uses a rolling 30 s clock for each 40 m effort.

Protocol

- The athlete starts 0.5 m behind the timing gate.
- The athlete's foot must touch the line at the end of the 20 m turning point. Failure to do so means the test is stopped, the athlete rests 5 min, and the test is repeated.
- During the recovery phase, it is essential that the athlete complete the 20 m recovery, going around the cone set 10 m behind the starting line.

Pre-Test

- The athlete starts 0.5 m behind the timing gate. On the tester's call, the athlete performs a maximal 40 (20 m shuttle) sprint.

- The athlete must ensure that a maximal effort is achieved. The tester is responsible for providing verbal encouragement.
- The time is recorded and the athlete is given an additional 5 min recovery before the RSA test begins. This score is used to ensure all sprints are of maximal effort.

RSA Test

- The athlete performs 6 repeated 40 m efforts on a rolling 30 s clock.
- The time taken for the sprint dictates the recovery time available within the rolling 30 s.
- All times should be recorded for the 6 sprints with verbal encouragement given throughout.
- For the active recovery, the athlete completes a 10 m shuttle (total 20 m) in the recovery period.

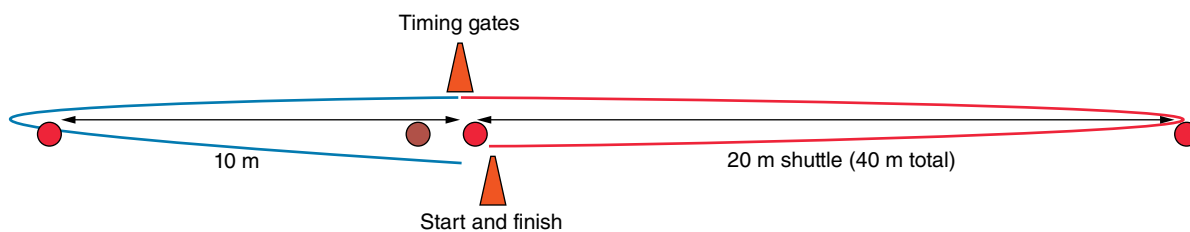


Figure 25.4 Setup for the Hamilton RSA protocol for field hockey players. Red line: sprint; blue line: active recovery.

training should be focused. The substitution component of field hockey allows us to maximize output and facilitate a higher intensity of work on the field. Players typically have multiple playing spells lasting 3 to 12 min (position dependent). Figures 25.5 and 25.6 represent in-house data collected by teams from over 60 international matches. Figure 25.5 shows the expected relationship between player work rate speed and spell length, while figure 25.6 similarly shows the correlation between work rate and total playing time during international field hockey matches. These data show the expected reduction in work rate speed as either shift length or total playing duration increases.

Clearly, having athletes play for longer durations results in lowered work rates, high-intensity actions and work output. Figure 25.7 highlights how the combination of physiological benchmarking and profiling can be incorporated to target the conditioning areas needed to develop field hockey athletes.

Targets of Physical Performance

The unique movement patterns required by field hockey players may be attributed (at least in part) to the right-handed-only sticks players use to

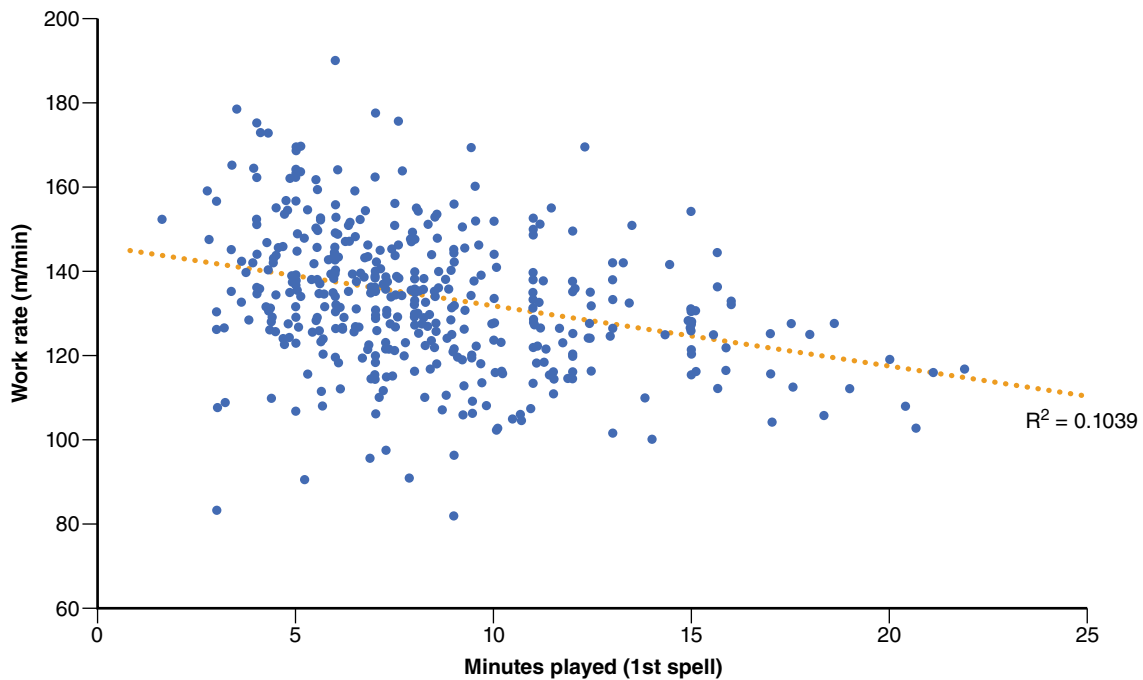


Figure 25.5 Relationship between first spell minutes played and work rate speed during international field hockey match play. Note the expected general decline in work rate speed with increased first spell duration.

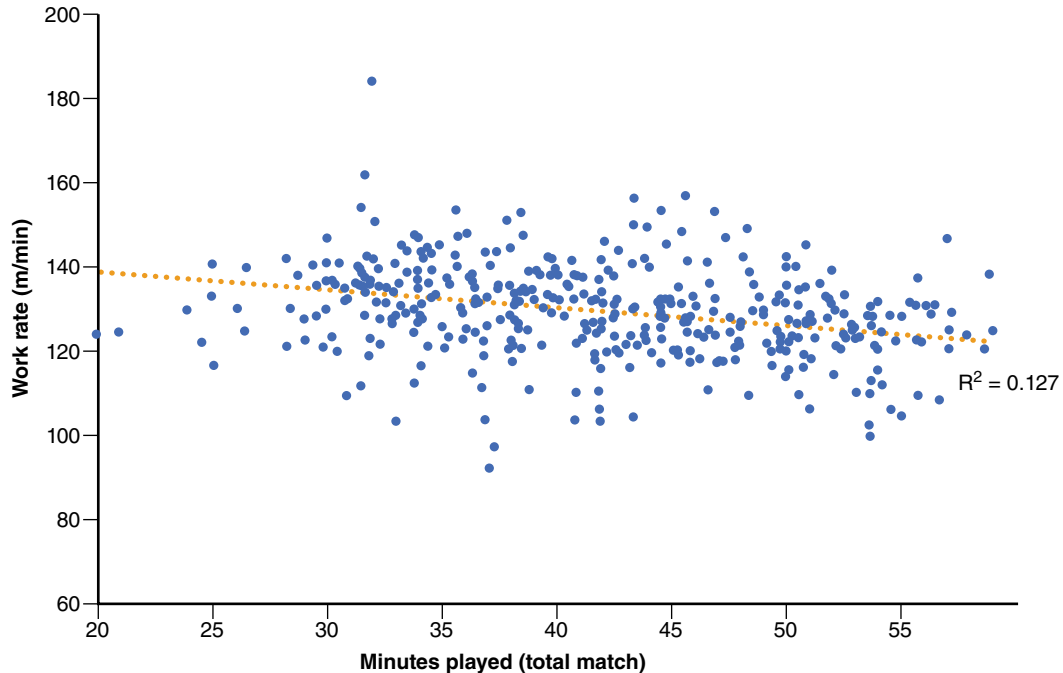


Figure 25.6 Relationship between duration of minutes played and work rate speed in international field hockey matches. Note the expected general decline in work rate speed with increased minutes played.

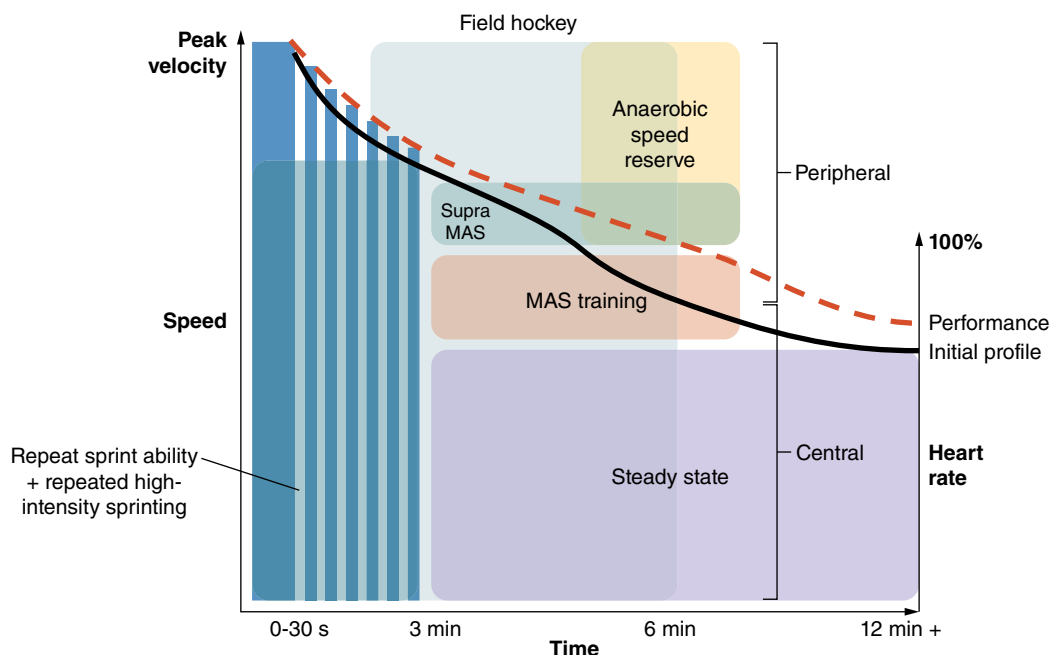


Figure 25.7 The Hamilton field hockey physiology profile used for benchmarking and training targets. MAS: maximal aerobic speed.

control the ball. Holding the stick with the left hand at the top and the right hand closer to the bottom half of the stick has led to a number of predetermined movement patterns associated with tackling, running with the ball, striking the ball, and free running without the ball. Inevitably, such skill sets can create unilateral asymmetries as players consistently become exposed to repetitive asymmetrical musculoskeletal loads. The left leg is often the preferred and functionally more proficient leg adopted during the forward lunge when tackling (left hand to the ground) and when striking the ball as a plant leg. High deceleration loads are placed on the left leg compared to the right, and programming will often need to reflect these demands to appropriately prepare the players. The highly explosive rotational actions performed in deep-flexed positions bring about other stresses as they pertain to muscle function around the hip and spine. Additionally, the sport requires these actions to be performed for 60 min while covering significant distances (5 to 9 km) up and down the field while enduring all the physical qualities needed to be a successful athlete in an evasion-based sport. At a club level, fixtures will take place once a week and weekly training is scheduled in such a manner as to help players recover early in the week, subject them to loading midweek, before tapering or unloading appropriately toward the end of the week and the next game. Internationally, the

demands on players are different again. In this environment, where the game is even faster (1), players are required to play on consecutive days and often over an 8 to 12 d period in the Olympic Games and World Cup, a competition schedule unlike any other. Understandably, preparation approaches between club and country commitments will differ substantially due to the constraints of the environment and the type of competition schedule being prepared for.

Time-motion studies reveal that athletes typically cover 5 to 9 km over the course of a game, with 12% to 26% of that time spent performing high-intensity running or sprinting. For the purpose of this chapter, high-intensity actions (HIA) have been assessed using GPS data and are classified as any running volume above 15 km/h (9 mph). GPS systems have become invaluable tools in more recent times for helping coaches and practitioners better quantify the work being completed during training and in competition. This numerical understanding of physical workload has assisted us to refine training methodologies, to manipulate training through drills and small-sided game strategies, as well as make position-specific modifications, work:rest interval adjustments, and understand a player's training load (see chapter 8). GPS technology allows us to set our own metrics to describe the relevant work being completed, including speed zones, number of accelerations, and types of accelerations (easy, hard, etc.). The

Table 25.2 Time-Motion Analysis and GPS Speed Zones (km/h) for Male and Female Hockey Players

	Speed km/h						
	Zone 1 standing	Zone 2 walking	Zone 3 jogging	Zone 4 running	Zone 5 striding	Zone 6 sprinting	Zone 7*
Male	0-0.6	0.6-6	6-11	11-15	15-19	19-21	21+
Female	0-0.6	0.6-6	6-11	11-15	15-19	19+	

* Zone 7 sprinting used only for men as an additional sprinting metric.

Table 25.3 Average Time-Motion Data for International Male and Female Hockey Players (120+ Games), 2009-2016

	Total distance (m)	Work rate (m/min)	HIA distance >15 km/h	% HIA	Average playing time (min)
Female	5700	128	1050	18	44
Male	6000	136	1240	19.5	44

Table 25.4 Average Positional Data, Including Total Distance, Work Rate, High-Speed Running, and Average Spell Length From International Field Hockey Play Across Forwards, Midfielders, and Defenders

Position	Total distance (m)	Work rate (m/min)	High-speed running (> 15 km/h)	Average spell length (min)
Forward	4544	130	1000	6.5
Midfielder	5511	128	1117	8.6
Defender	5783	115	605	14.7

downside of such flexibility with this technology arises from the difficulty that then transpires when trying to compare data sets between teams and across sports. Table 25.2 highlights the speed zones used by the Great Britain and US Olympic field hockey teams during the 2009 to 2016 quadrennial cycles. As a means to understand and quantify training and game loads for athletes, time-motion data (GPS) was used to quantify exactly how far athletes were running at different speeds during a game. These zones were chosen to reflect game impact and describe the type of work being completed.

As shown in figure 25.2, the aerobically demanding, intermittent anaerobic nature of field hockey is not dissimilar to that of many sports detailed in this book, including soccer or Australian rules football (9). Therefore, a well-developed aerobic system is integral to field hockey performance, not only for the consistent active nature of field hockey but also for the metabolic recovery needed between high-

intensity efforts. While there is consensus on the overall physical demands of field hockey, there is some variability and differences across genders and positions, as highlighted in table 25.3.

Outfield Players

As previously mentioned, there is a wide range of physical performance requirements and countless high-intensity actions superimposed on the aerobically demanding game that is field hockey. Developing the aerobic system, a favorable body composition, and high relative strength levels is paramount for players to be able to sustain high-velocity actions and recover effectively between the multiple bouts of low-to-ground activities that can shape the outcome of a game. Outfield players can be broken down into the positional groups of forwards, midfielders, and defenders (or screens), with positional data shown in table 25.4.

Goalkeepers

Goalkeepers have the most unique demands of any position in field hockey. For starters, keepers wear far more equipment than any field player, with large leg guards typically measuring 24 to 28 in. tall by 10 to 11 in. wide, padded shoes called *kickers*, large blocking gloves on each hand, a chest protector, padded pants or girdles, a helmet with full face protection, and a specialized goalkeeping stick. Furthermore, the physical requirements are substantially different from those of the outfield players.

There are many physical attributes thought to be important for success in the position. A keeper should be kinesthetically aware of her limb positions, possess high levels of power-to-weight due to the additional equipment loads she must bear, have strong lateral and linear speeds, alongside high rates of force development (RFD) as they pertain to mitigating movement lag toward the ball or players as required. Reactive qualities highly depend on high relative strength levels, RFD, and neural drive. Goalkeepers must develop and embed these characteristics within both lower- and upper-limb movements.

The ability of a goalkeeper to track game play and position in or around the goal in such a manner as to manipulate an advantage in the goal space is critical to making scoring a goal as difficult as possible for the attacker. With this skill comes the necessity to perform both linear and lateral movements, both of which need to be performed in an optimal athletic posture that allows key reactive movements to be performed.

Often during shot attempts on goal, goalkeepers will commit to a fully extended ground position to block the shot attempt. Unfortunately, such positions can leave the goalkeeper and goal vulnerable should a deflection occur. It is crucial therefore that a goalkeeper possess sufficient strength and power qualities to react and return to her feet as quickly as possible. This ability requires excellent overall body strength, power, and limb coordination. Additionally, shoulder strength and arthrokinematics (movement at joint surfaces) are thought to be important for goalkeeper success.

Impact of Game Format Change: Halves Versus Quarters

In 2014, the game format at the international level changed substantially, shifting from two halves of 35 min to a four-quarter format involving 4 × 15 min

periods of faster play (1). Play is separated by 2 min rest periods between quarters 1 and 2 and between quarters 3 and 4, with a 10 min halftime stoppage between quarters 2 and 3. This format was introduced to complement TV formats at the Olympic Games, in the hopes of improving viewer interest by providing an even faster and more explosive brand of international field hockey. Figure 25.8 shows a GPS data set comparing an international team across the transition from one international season playing halves into the new international quarter format. These data highlight the substantial impact of the format change as it relates to the physical demands of the athletes, with 5% to 10% increases in overall work rate and high-intensity work rates generally shown across all outfield playing groups.

It is apparent from this data that the change in game format (halves to quarters) has had a meaningful impact on the positional lines at the international level (figure 25.8). With average total playing time being reduced to accommodate the 10 min game length reduction, all playing lines have had reductions in their average playing spell length times. Additionally, this reduction in field playing time has facilitated an increase in work rate (meters per minute) and HIA (distance covered over 15 km/h) in the new 60 min game compared to its 70 min counterpart. While there is no observable change in total distance covered per game, players must work harder to cover a similar total distance in a shorter time. This has implications as to how teams should prepare and train players, and hence the importance of prescribing the most appropriate HIIT weapon for hitting these aforementioned targets.

Key Weapons, Manipulations, and Surveillance Tools

In this section, I present the different HIIT weapons used in field hockey, their manipulations, along with ways we can use surveillance to help keep us on track throughout an international field hockey season.

HIIT Weapons

Different times within a training cycle or season call for bespoke changes in training priorities. Understanding how and when to make these adjustments is paramount to running a successful program. The ultimate goal must be to develop athletes who are capable of withstanding the demands of the

Position	Game type	Total distance	Work rate m/min	Total distance sprinting	% HIA	Total playing minutes	Average spell
Defender	Half	5783	115	605	10.8	51	14.7
Defender	Quarter	5978	122	651	11.2	48	10.5
Percentage difference		3	6	8	4	-6	-29

Position	Game type	Total distance	Work rate m/min	Total distance sprinting	% HIA	Total playing minutes	Average spell
Forwards	Half	4544	130	1000	21.7	36	6.5
Forwards	Quarter	4622	135	1174	23.5	32	5.5
Percentage difference		2	4	17	8	-11	-15

Position	Game type	Total distance	Work rate m/min	Total distance sprinting	% HIA	Total playing minutes	Average spell
Midfields	Half	5511	128	1117	20.1	42.6	8.6
Midfields	Quarter	5528	134	1231	22.2	41.4	6.7
Percentage difference		0	5	10	10	-3	-22

Position	Game type	Total distance	Work rate m/min	Total distance sprinting	% HIA	Total playing minutes	Average spell
Screens	Half	5918	124	885	15.3	48	10.9
Screens	Quarter	6003	131	1095	18.2	45	7.7
Percentage difference		1	6	24	19	-6	-29

Figure 25.8 Half versus quarter comparison in key GPS markers across international field hockey time-motion data for all positional groups.

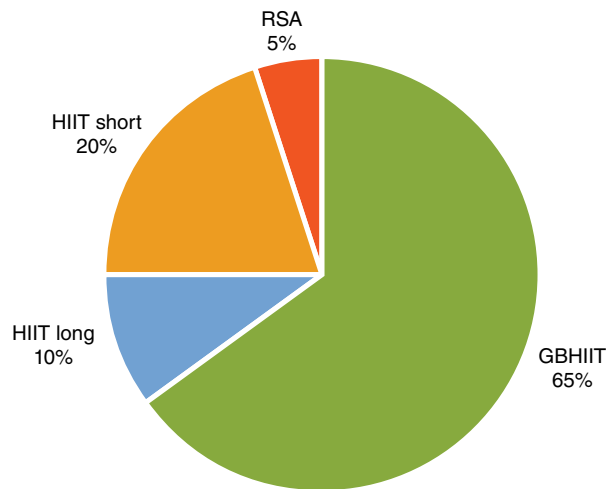


Figure 25.9 Percentage of the different HIIT formats (weapons) used throughout the annual season in elite field hockey: GBHIIT 65%, HIIT with long intervals 10%, HIIT with short intervals 20%, RSA 5%.

sport, allowing them to perform to their potential. Ultimately, we need to develop athletes who are resilient to the asymmetrical stresses of the sport and the demands of a competitive schedule. Using HIIT appropriately is key in this regard. The percentage breakdown of the different HIIT weapons pre-

dominantly used to prepare field hockey players is shown in figure 25.9.

Manipulations of Interval Training Variables

The initial focus of any training program will be dictated by the phase of the training year. Being in season or out of season will heavily determine the focus of the physical qualities we address and the structure of training to prioritize the desired outcomes. Out of season for a club team often refers to any period outside of the competitive league fixture list. However, for the international teams the out-of-season tag is slightly more fluid in order to accommodate changing annual schedules and rotation into major events, alongside the different locations and qualification processes. The international fixtures typically occur from October to August, although there may be opportunities for breaks between international competitions. Generally, there is a break from all international fixtures during September and October, and often this follows the last major event in the international calendar (World Championships, Continental Championships, or the Olympic Games).

Immediately postseason offers a great opportunity for regeneration of the body and mind following extended exposures to the demands of the game. Like anything, however, a balanced approach to

recovery and training is paramount for continued success. Returning too soon to training can lead to the resurgence of soft tissue issues, if injuries aren't fully recovered, whereas returning too late can result in inappropriate loading relative to the level of resiliency. As a result of this, and in order to progress the physical development of the athletes, it is paramount that physical demands accumulated during the season are not altogether lost due to inactivity. Neuromuscular aspects (type 2, 4, and 5 targets), including changes of direction and the high mechanical stress associated with low-to-ground actions, evasive activities, and the stop-start nature of the running patterns, mean that these need to be accommodated for during training when out of season. Repeated sprint activities, shuttles, and interval-based work should be present, as should customized

agility drills that allow athletes to train with their stick in hand. Adding the stick provides a training stimulus that facilitates additional loads and more-specific movement patterns. Similar to a monkey using its tail for balance, a stick in the hand will allow for sport-specific foot patterning and loading patterns, thus improving musculoskeletal tissue tolerance levels for future hockey demands. A 6 wk out-of-season RSA drill program is shown in table 25.5 and figure 25.10, while the Hamilton RSA protocol is described earlier in the chapter and in figure 25.4.

The positive side effect of these types of RSA and shuttle activities is that they facilitate higher neuromuscular stress (type 2, 4, and 5 targets) and develop the tissue tolerance levels needed to prepare athletes for the demands of on-field activities when

Table 25.5 Six Week Out-of-Season Repeated Sprint Ability (RSA) Drills for Field Hockey

Session	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
Dynamic warm-up	15 exercises × 15 m (16 yd)	15 exercises × 15 m	15 exercises × 15 m	15 exercises × 15 m	15 exercises × 15 m	15 exercises × 15 m
RSA (40 m) always on a rolling 30 s cycle			3 × 6 sprints (2.5 min rest between sets)	3 × 7 sprints (3 min rest between sets)	2 × 6 sprints (1 drill), 2 × 6 sprints (another drill), 5 min between drills, 3 min between sets	2 × 6 sprints (1 drill), 2 × 6 sprints (another drill), 5 min between drills, 2.5 min rest between sets

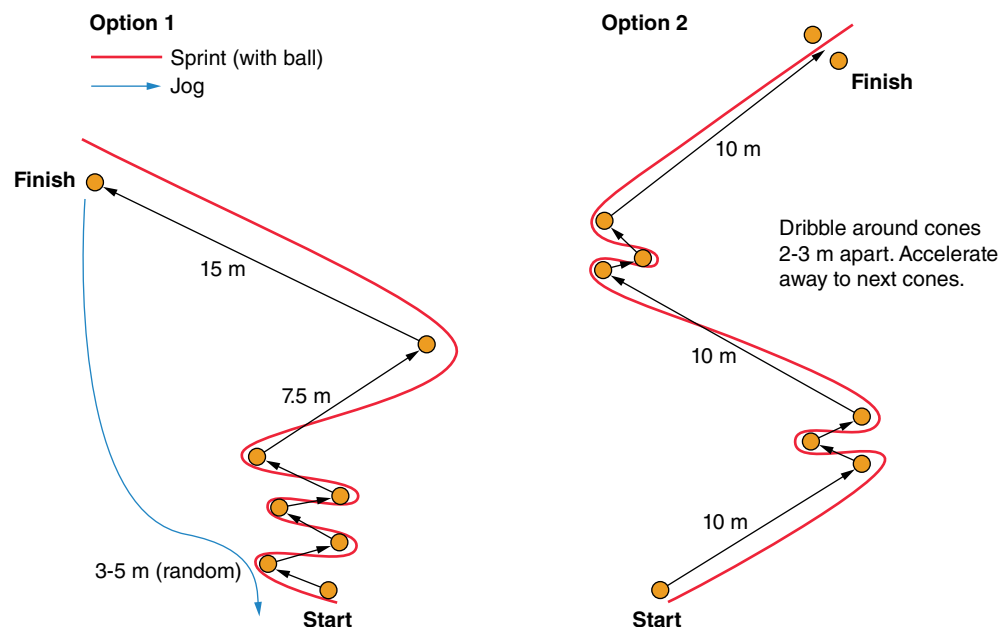


Figure 25.10 With stick and ball repeated sprint ability (RSA) drill details.

Table 25.6 Six Week Out-of-Season Field Hockey Speed and Conditioning Plan

Week	Monday	Tuesday	Wednesday
1			Conditioning 4 mile (6.4 km) run at self-selected pace Focus: central adaptation
2	Recovery: bike 30 min at 60 to 80 rpm Low resistance Type 1		Conditioning 4 mile run at target sub 30 min Focus: central adaptation Type 1
3	Speed session	Conditioning treadmill 5×3 min at 85% (30-15 V _{IFT}), 3 min rest between sets Focus: central adaptation Type 3	
4	Speed session	Conditioning treadmill 5×3 min at 88% (30-15 V _{IFT}), 3 min rest between sets Focus: central adaptation Type 3	Speed session
5		Conditioning treadmill 6×3 min at 85% (30-15 V _{IFT}), 3 min rest between sets Focus: central adaptation Type 3	
6	Speed session	Conditioning 4 mile run at sub 28 min Type 3	

out of season. However, the negative aspect or challenge to these activities is that they facilitate higher neuromuscular stress. Therefore it is critical to have a well thought out plan and weekly structure to ensure the volume of work completed is actualized in a manner favorable to long-term athlete development. Understanding the recovery needs following any training stress is also important and will depend on the system trained, load of the activity performed (chapter 8), training history, and individual aspects. We know that a *physical poke* results in some kind of metabolic or neuromuscular response and the degree of that response depends on the individual athlete, their fitness level and prior training history, along with the intensity, duration, and mode of activity (see chapter 9).

During the off-season and preseason phases, we should look to take direction from what is learned from training load monitoring in season and around the constraints of competition. By quantifying and periodizing these training loads, we can strategically prescribe training that accurately replicates the in-season load demands. Replicating exactly the in-season training load is not the solution. However, progressive exposure to competition load levels is more the goal. Using the HIIT weapons appropriately is key in this regard. Table 25.6 shows an example of a 6 wk out-of-season field hockey speed and conditioning plan, while figure 25.11 provides conditioning drill details.

Maximal aerobic speed (MAS) prescriptions, or long intervals (greater than 3 min exposures of

Week	Thursday	Friday	Saturday	Sunday
1				Conditioning treadmill: 4×4 at 82% (30-15 vIFT) Focus: central adaptation Type 3
2	Speed session		Conditioning hockey match or 3×10 15:15 shuttle at 95% (30-15 vIFT), 3 min rest between sets Focus: introduce COD and peripheral adaptation Type 4	
3	Conditioning repeat sprint (see options in figure 25.11) Focus: peripheral adaptation Type 5		Conditioning hockey match 3×14/12/12 at 95% 15:15 straight, 3 min rest between sets Focus: peripheral adaptation Type 4	
4	Conditioning repeat sprint (see options in figure 25.11) Focus: peripheral adaptation Type 5		Conditioning 3×12 at 100% 15:15 straight, 3 min rest between sets Focus: peripheral adaptation Type 4	
5	Conditioning repeat sprint (see options in figure 25.11) Focus: peripheral adaptation Type 5		Conditioning 3×14 at 95% 10:10 straight, 3 min rest between sets Focus: peripheral adaptation Type 1	
6	Conditioning repeat sprint (see options in figure 25.11) Focus: peripheral adaptation Type 5		Conditioning 3×12 at 100% 10:10 straight, 3 min rest between sets Focus: peripheral adaptation Type 1	

running), are an excellent training methodology to elicit a physiological poke guaranteed to promote adaptation. This type of training prescription is multifaceted when it comes to effectiveness, as it is simple to prescribe and individualize and can be performed in multiple training environments (gym treadmill, courts, fields, tracks, etc.). It offers an appropriate volume and intensity of training with a high physiological adaptation response to facilitate central adaptations (type 3 and 4 targets), while mitigating the excessive running, sometimes referred to as junk miles, associated with long slow runs (LSR) or traditional fartlek approaches. Both LSR and fartlek work have their place in off-season training, but more as a means of amplifying running volume in the week when accurate prescriptions of distances and

time are not possible, such as during a vacation where facilities may be limited.

Periods of training at or near $\dot{V}O_2\text{max}$ for sustained periods of time will provide the necessary training stimulus to promote central and peripheral adaptations within the aerobic system (chapter 3). Therefore the maximal aerobic speed (MAS) sessions described across a 4 wk block of training are an excellent means of ensuring that sufficient training intensity and duration are achieved. In this weekly structure, we have a balanced approach to the management and development of the aerobic system, while controlling for the high neuromuscular demand of small-sided game (SSG) and hockey match play throughout the week. Straight-line MAS is the preferred training methodology during the

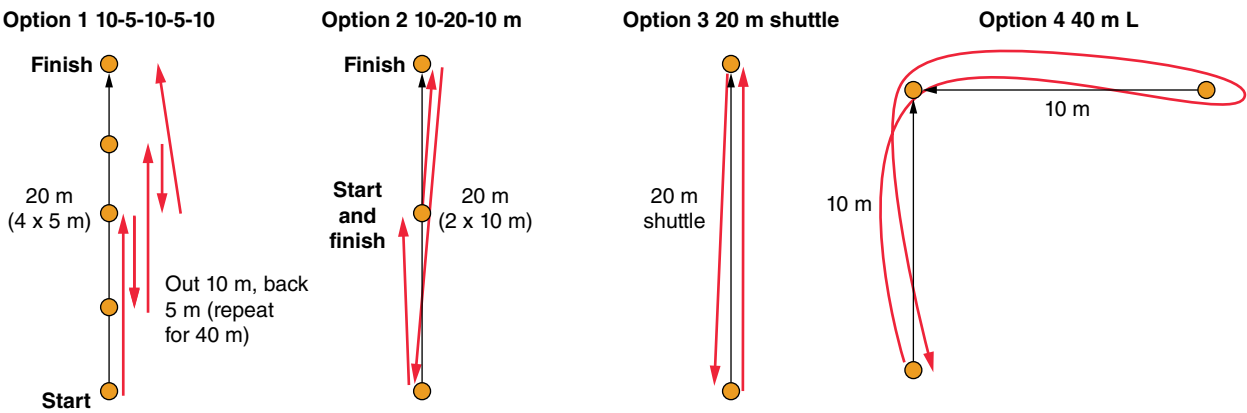


Figure 25.11 Without stick-and-ball drills.

Table 25.7 Seasonal Prescription Loads for MAS (Using 30-15V_{IFT} Values for Prescription) During a Training Week

	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
MAS 1 (straight)	3 × 12 at 95% 15:15	3 × 12/10/10 at 100% 15:15	3 × 12/10/8 at 103% 15:15	3 × 14 at 95% 10:10	3 × 12 at 100% 10:10	3 × 12/10/10 at 105% 10:10
HIIT type	Type 2	Type 3	Type 4	Type 2	Type 3	Type 4
MAS 2 (shuttle)	3 × 9 at 95% 15:15 shuttle	3 × 8 at 98% 15:15 shuttle	3 × 7 at 100% 15:15 shuttle	3 × 9 at 95% 10:10 shuttle	3 × 8 at 100% 10:10 shuttle	3 × 8/8/6 at 102% 10:10 shuttle
HIIT type	Type 2	Type 4	Type 5	Type 4	Type 5	Type 5

in-season training week. This type of straight-line prescription allows for controlled exposure to the training intensities required to elicit an appropriate dose of aerobic responses, while mitigating the unnecessary mileage and high neuromuscular loads associated with the mechanical loading of musculature during shuttles and changes of direction. A 6 wk prescription of training using 30-15V_{IFT} values for prescription is shown in table 25.7.

During the training week in season, I prioritize 1 or 2 additional MAS sessions to help facilitate improvements to this system. Game-based HIIT or small-sided games (SSGs) are fantastic for developing myriad physical attributes (speed, COD, lactate profiling, anaerobic and aerobic processes) through the manipulation of player numbers, pitch size, playing duration, and rest. However, despite the high level of variety possible in game structure, we are unable to guarantee total development to all players uniformly during these sessions. SSGs can offer an opportunity for individuals to hide, control effort lengths, and even limit their development and improvement opportunities.

As the off-season progresses and the season approaches, the run volume (distance/time) prescription within the repetitions is lowered. This small tweak in repetition volume allows for a higher number of total repetitions to be performed, along with a larger volume of accelerations and decelerations and higher running speeds. This change in prescription allows for transition from the centrally focused adaptations more toward the peripherally focused ones (figure 25.7; see also chapter 3). This is a training adaptation unique to the prescription and a key quality that will help underpin high-intensity actions throughout a game. I recommend performing such long intervals 2 to 4 times a week, separated by 24 to 48 h of recovery.

Load Surveillance and Monitoring Tools

Successful training prescription is most effective when we adopt an athlete-centered approach and develop our understanding of both training load

(chapter 8) and the individual athlete response (chapter 9). It is important that we create a training environment that is rich in information to remove subjectivity around our loading and unloading strategies. The ultimate goal for athlete management has to be having our athletes arrive at their competitive peak in a predictable way. This requires a thorough needs analysis of the athlete and an integrated approach to assessing the components that will positively impact on performance by developing unique and bespoke monitoring systems.

As with any athlete who strives to improve performance, there is often a systematic increase in training frequency, volume, and intensity of training. This increase in training load is often successful and represented as a performance improvement. However, athletes will also experience varying severe levels of fatigue during such challenges, and since performance and training can be compromised during such periods, it is paramount that effective monitoring strategies are put in place. It is the role of the support staff to effectively monitor and assess fatigue to facilitate optimal recovery and prevent abnormal fatigue states or overtraining from occurring (see chapter 7).

As described in chapters 8 and 9, the physical stress that forms the training load creates a physiological response, and the extent of the response will vary depending on the individual and their reaction to the intensity, volume, and type of activity performed. A performance program rich in information will allow you to more effectively understand the individuals on the team. Monitoring should involve a set of activities designed to help establish the acute training status of the athlete, so that appropriate and timely training adjustments can take place that ensure ongoing performance targets are attained. Essentially, the outcome from a monitoring system should be to produce a training environment that facilitates the greatest amount of development and ensures optimal performance around the competition program. Once we have the information, we can be well situated to make accurate and timely decisions around recovery interventions, physical development, and competition strategies.

Training Status Surveillance and Monitoring Tools

The primary training status surveillance and monitoring tools we use in field hockey include GPS for

external load and session rating of perceived exertion (RPE) for internal load, with wellness questionnaires and drop jumps to assess the response to loading.

GPS Systems (External Training Load)

As discussed throughout the chapter, GPS systems are used extensively by international hockey squads to measure external training load. GPS systems have become invaluable tools for helping coaches and practitioners better quantify the work being completed during training and in competition. This numerical understanding of physical workload has helped us refine training methodologies, manipulate training through drills and SSG strategies, and make position-specific modifications and work:rest interval adjustments, and understand a player's training load (see chapter 8). As a means to understand and quantify training and game loads for athletes, time-motion data (GPS) can be used to quantify exactly how far athletes are running at different speeds during a game. Zones can be chosen to reflect game impact and describe the type of work being completed (see tables 25.2 to 25.4 and figure 25.8).

Session RPE (Internal Load)

Session RPE is an excellent way for the field hockey coach and athlete to monitor training intensity without equipment (see chapter 8). Numerous studies have shown that session RPE is an excellent way to assess globally an athlete's perception of internal training load (4) and this has proven effective for a variety of resistance and conditioning session types (11). RPE is extremely useful in the context of field hockey player monitoring across the large variety of training modes players undergo during a typical training week. An example of a session RPE load capture across 4 wk is shown in figure 25.12.

Wellness Questionnaires

It is generally accepted that excessive training stress can be associated with psychological disturbances and changes in disposition as well as marked changes to both cognitive and somatic markers in the body (chapter 7). In order to address such factors, we use wellness questionnaires to assess daily wellness scores and extensively monitor these varying manifestations of stress. It is important to realize that for a wellness tool to be a truly effective indicator of physical performance, there is a need for it to be used in conjunction with other physical performance

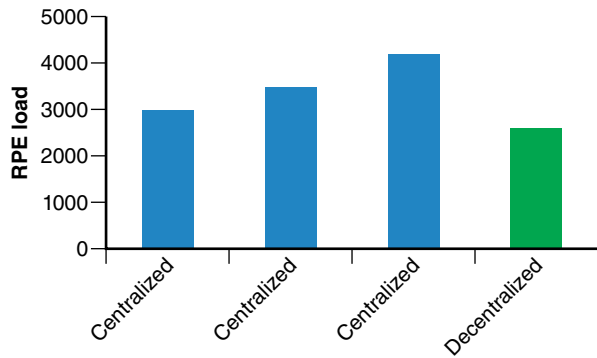


Figure 25.12 A four-week linear model of loading measured using session rating of perceived exertion (RPE).

tests that offer insight into the mechanistic changes that may be impacting the athlete.

Neuromuscular Assessment

When considering the impact of field hockey on an athlete and trying to understand the implication of training and competition on performance, it is important to look beyond the metabolic cost of energy systems and to also consider the mechanical cost of low-to-ground actions, high decelerations, and multiple changes of direction. Fatigue will manifest in different facets of the body's functionality, and an area often overlooked due to feasibility of assessment and resourcing is the neuromuscular system (chapters 3, 4, and 5). A compromised neuromuscular system has an effect on force, velocity, and power qualities and ultimately performance potential (3). Assessment of neuromuscular fatigue, including drop jumps and the reactive strength index, is described in chapters 3 and 8.

Strategies for Structuring the Training Program Using HIIT

I use a number of strategies in field hockey to structure the training program and HIIT prescription, which must be made alongside the context of the program, and its controllable and uncontrollable factors.

Controllable and Uncontrollable Factors

Time availability for training specific physical qualities can become limited during the season and

during the training week. It is fundamental that we understand what truly underpins and impacts success in our sports. Experience, mastery, and exposure to the event are key to long-term success within our chosen fields. Learning from situational play, repeated exposure to game-specific situations, and developing mastery of our skills form the backbone philosophy of our program. Our main goal then is to give our international athletes back to our club coaches as often as we can. Nevertheless, having the ability to compromise is imperative for success, and the ability to pick the training tools to invoke the largest and most appropriate training response in a timely fashion is what separates the great teams from the good ones.

Program Structure and Progression

Due to the length of an international season and the extended preparation time required, it is important to build an intelligent program that provides resiliency in the athletes through appropriately progressed training loads and adequate recovery. With an international field hockey program that is often full time and centralized (athletes train out of the same facility), a 4 wk linear periodization model has been proven to be an effective means by which athletes may achieve this outcome (i.e., figure 25.12). The 4 wk linear model is structured in such a way as to facilitate an increase in hockey training frequency and volume each week. The program is built around a 4 wk mesocycle consisting of 3 wk of progressive hockey and physical development loading and 1 wk of unloading when no mandatory team hockey sessions are completed. Instead, physical qualities are targeted and off-feet conditioning becomes a priority. This 4 wk loading pattern facilitates an accumulation of hockey-specific training load, while also allowing for a regeneration week and a hockey-specific stress unloading period from the asymmetric demands of the game.

The advantage of this type of periodized model is that it develops most athletes' ability to tolerate multiple training sessions in a day and across consecutive days of hockey match play, while establishing timely recovery windows to support the regeneration of tissue from hockey-specific overload. This model involving 3 wk of hockey load complements and appropriately prepares the players for the demands of tournament hockey, where they are typically required

Sample periodized plan

Review

Microcycle week

Domestic leagues and test matches

Contracted training weeks

Training plan intensity

General thoughts

Training block focus

Competition

Strength and power

Speed, agility and biomechanics

Metabolic conditioning

Focus

July

August

September

October

November

December

January

February

March

April

May

June

Week 1

Week 2

Week 3

Week 4

Week 5

Week 6

Week 7

Week 8

Week 9

Week 10

Week 11

Week 12

Week 13

Week 14

Week 15

Week 16

Week 17

Week 18

Week 19

Week 20

Week 21

Week 22

Week 23

Week 24

Week 25

Week 26

Week 27

Week 28

Week 29

Week 30

Week 31

Week 32

Week 33

Week 34

Week 35

Week 36

Week 37

Week 38

Week 39

Week 40

Week 41

Week 42

Week 43

Week 44

Week 45

Week 46

Week 47

Week 48

Week 49

Test

Camp (away)

Camp (home)

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home

Pre season

Competition

Overseas

Home</

Figure 25.13 An example of a yearly macrocycle training plan in an international field hockey team.

Training phase	Comp schedule	Week	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday	Gym	Cond	Rob	Sp&ag	Reps	Gym intensity	Physical load 1 low 5-very high
Physical preparation ii	Regeneration	35	Rest	Bike recovery	Bike-interval	Circuit	Conditioning	Rest	Rest							
			Rest	Upper body	Robustness	Rest	Rest	Rest	Rest	2	2	1	0	12	55	1
	Centralized	34	Gym	Hockey	Gym-upper	Hockey	Gym	Conditioning	Rest	3	2	1	1	8	65	3
			Hockey	Hockey	Rest	Hockey	Rest	Rest	Rest							
	Centralized	33	Gym	Hockey	Gym-upper	Hockey	Gym	Conditioning	Rest	3	2	1	1	6	79	4
			Hockey	Hockey	Rest	Hockey	Rest	Rest	Rest							
	Centralized	32	Gym-testing	Hockey	Gym-upper	Hockey	Gym	Conditioning	Rest	3	2	1	1	3	85	5
			Hockey	Hockey	Rest	Hockey	Rest	Rest	Rest							
	Regeneration	31	Bike recovery	Gym-speed	Bike-interval	Gym	Speed* conditioning	Rest	Rest	3	4	0	2	5	65	2
			Gym-upper	Rest	Mobility	Rest	Rest	Rest	Rest							
	Centralized	30	30-15 + speed	RSA	Gym	Hockey	Hockey	Gym	Rest	3	2	1	1	5	72	3
			Gym-upper	Hockey	Rest	Hockey	Rest	Rest	Rest							

Figure 25.14 A sample mesocycle bimonthly training plan in an international field hockey team.

Table 25.8 Sample Microcycle (One Week) Training Plan of an International Field Hockey Team

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Hockey	Hockey	Gym	Hockey	Hybrid: gym and hockey	Conditioning: MAS 2	OFF
Gym	Hockey + MAS 1		Hockey + SSG			OFF

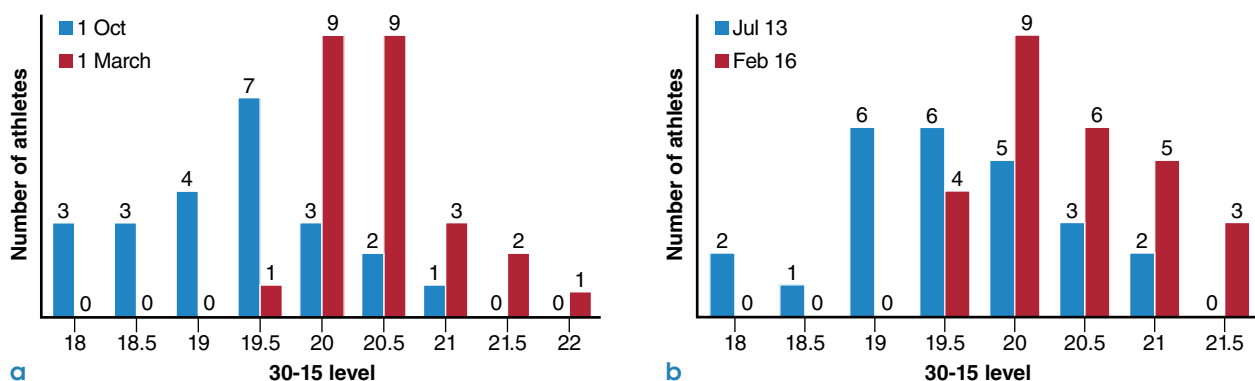


Figure 25.15 The impact of MAS prescription and changes in 30-15 IFT performance in two international field hockey teams across two quadrennial cycles, (a) 2012 and (b) 2016 Games. Pre-value levels in blue, post-value levels in red, with numbers representing the number of team members achieving that level.

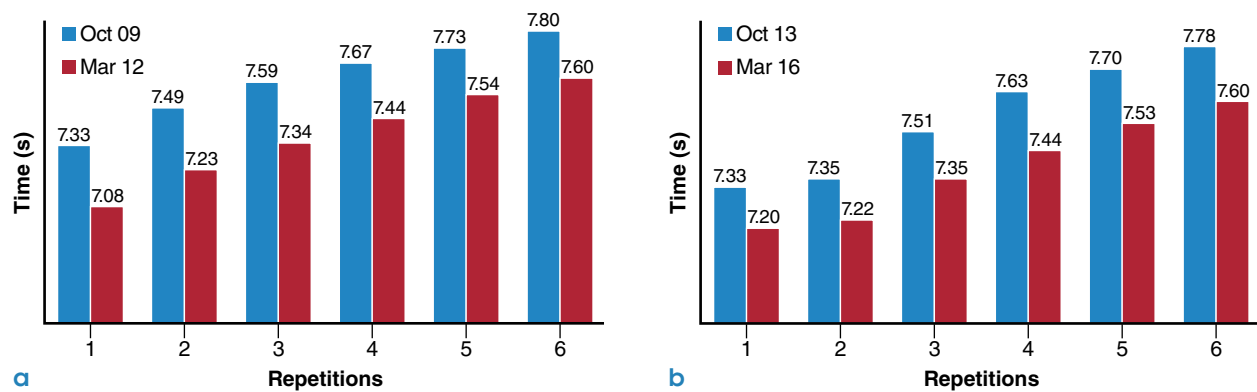


Figure 25.16 The impact of repeated sprint ability (RSA) prescription on changes in RSA performance in two international field hockey teams across two quadrennial cycles: (a) 2012 and (b) 2016 Games. Pre-value levels in blue, post-value levels in red, with numbers representing the number of 40 m sprint repetitions achieved.

Table 25.9 Physical and Performance Markers in Two Separate Teams Across Two Olympic Cycles

Measures include body weight, relative back squat and bench press performance, as well as 5 m, 10 m, mean repeated sprint ability performance(s), and 30-15 intermittent fitness test (30-15) level.

Team	Date	Body weight (kg)	Relative back squat	Relative bench press	5 m (s)	10 m (s)	Mean RSA 6×40 m shuttle (s)	30-15 level (km/h)
2012 Olympic team	Late 2009	65.6	1.29	0.68	1.11	1.85	7.87	19.2
	Early 2010	67.3	1.58	0.80	1.05	1.82	7.62	20.1
	Late 2010	64.5	1.35	0.76	1.07	1.85	7.68	19.4
	Early 2011	64.4	1.52	0.86	1.00	1.78	7.54	19.7
	Late 2011	63.2	1.55	0.89	1.07	1.81	7.54	20.2
	Early 2012	63.2	1.59	0.95	1.00	1.77	7.40	20.5
2016 Olympic team	Late 2013	61.0	1.55	0.91	1.02	1.79	7.54	19.4
	Early 2014	60.7	1.58	0.95	1.01	1.77	7.47	20.0
	Early 2015	60.4	1.65	0.90	0.97	1.76	7.43	20.2
	Late 2015	60.0	1.76	0.99	0.97	1.72	7.37	20.3
	Early 2016	59.8	1.70	0.96	0.97	1.74	7.42	20.4

to compete on consecutive days. In fact, during the major events of international field hockey—Champions Trophy, Continental Cups, World Cups, and the Olympic Games—teams are expected to complete 7 games in 12 d. Ultimately, this model of 3 wk on and 1 wk off will allow for the development of athlete robustness and resilience alongside a training structure that promotes longevity.

Specifically, we look to develop three qualities that underpin field hockey performance in our athletes, including:

1. **Durability:** ability to withstand the daily wear and tear that a prolonged central program delivers, while avoiding the potential injury risks associated with repetitive movements.
2. **Robustness:** development of strong and tough tissue that has adapted to the demands of hockey-specific overload and can tolerate a high frequency of repetitive movements.
3. **Recoverability:** an innate resilience to fatigue, with efficient movement skill and an ability to maintain postures and positions for prolonged periods.

Having clear goals and training objectives established throughout a training year is important, but understanding how to structure the training week

(microcycle) will form the foundation and key building block needed to make the goals achievable.

Sample Training Programs

Macrocycle, mesocycle, and microcycle examples of international field hockey teams are shown in figures 25.13 and 25.14 and table 25.8, respectively.

Longitudinal Training Effects

Figure 25.15 represents the impact of MAS prescription and positive changes in 30-15 IFT performance in two international field hockey teams across two quadrennial cycles, while figure 25.16 shows related outcomes of RSA training and Hamilton RSA performance along a similar time course in the same teams. Table 25.9 shows the longitudinal physical and performance markers measured across two Olympic cycles in two separate teams.

Acknowledgments

I would like to thank Bobby Lucas for his assistance with data collection and analysis. The experiences had with players and staff involved throughout the two Olympic campaigns described herein are also greatly valued.

CASE STUDY

It was a brisk Tuesday afternoon at the Olympic park in Stratford, London. Our Team GB hockey team was still 5 mo from the Olympic Games, but as host nation, we had the ability to gain brief access to the Olympic venue. This was an edge we believed we couldn't pass up, even if it meant we would have to endure a 2 h bus journey through London traffic, countless nights in hotels, eating our lunch pitch side, and meeting in cold porta-cabins sitting on the floor. No stadium was built at this point—just the pink and blue turfs, surrounded by scaffolding and heavy-duty construction trucks. While not the ideal training venue, the opportunity to increase familiarity with the Olympic turf, see our Olympic pitch develop, and be a part of the process could offer us the advantage we were after. As the rewards of playing at the unfinished Olympic turf far outweighed the logistical and administrative difficulties of getting there, we made systematic trips to the Olympic park over the 12 mo lead-in period to the Games, each time developing familiarity with the turf.

We felt our preparations for the London Games were on track and going well. We had witnessed some exciting changes within the group physically, both on and off the field. Our world ranking status was steadily improving since 2010, and we had medaled at all but one major event since the team centralized at Bisham Abbey (October 2009). Our team was now reaping the rewards of our preparation, ensuring we left nothing on the field during the quadrennial plan in our search of a podium finish in our home Olympic Games. Visiting the partially constructed Olympic park regularly was one of the important decisions.

On this particular Tuesday afternoon at the Olympic park, training was to have a surprise element to it, a calculated and planned element that none of the athletes were aware of. The content of the session had been devised 1 wk earlier as a means of rectifying a small glitch in the system. The team was coming off a couple of favorable training weeks and into the decentralized week, during which athletes had no national hockey team commitments and a light centralized training week (figure 25.12). However, on this particular occasion, we were unexpectedly not seeing as positive a response in the group's recovery as we were more accustomed to seeing. Players were tired, complaining of a lack of rest in the session content, as well as with the session's difficulty levels. Warm-ups were taking longer, and the staff had to work to motivate the team in exercises, and post-training conditioning runs were perceived as difficult. As a staff, we did what most coaches and support staff would do. We looked inward on ourselves and asked the obvious questions:

- Are we doing too much?
- Are we overtraining them?
- Do we have the timing all wrong?

But our performance metrics told us differently. By taking ourselves from Google Street View to Google Earth perspective, the disconnect was clear. There was no apparent reason for excessive fatigue. Perhaps it wasn't about overtraining, but more about the monotony of training. Could the repetitive nature of our training program, which requires athletes to rinse and repeat every 4 wk, have actually created a false fatigue response, manifesting more as mental, not physical, fatigue? Indeed, we too as staff had similar feelings.

But being 5 mo out from the Games was not the time to take the foot off the gas and risk losing the adaptations we had worked so hard to create. So instead, we decided that this was exactly the time to throw the rope around ourselves as a team, to refocus on our purpose, as well as the goals, commitment, and promise we had set as a team in October 2009. The Tuesday would provide the opportunity to take this group of aspiring Olympians to a place they had never been before. It would be an opportunity to show themselves as a group just how far they had come in their journey, and the level of robustness and resilience they had developed as a team. Fundamentally, the purpose of the session was to help the athletes reset their understanding of what *hard* actually is. While we know training monotony is undesirable from an injury perspective, it can also create a numbness or lack of feel to

(continued)

CASE STUDY (CONTINUED)

training load. This following planned exposure would potentially help reset the monotony cycle. At least that was our belief.

The Tuesday afternoon session at the Olympic pitch contained the usual content, finishing with 3 v 2 SSG toward the goal. The 2 h on the pitch was now up, and the team looked to be wrapping up the session, beginning to debrief with coaches. As was normal during this stage of the sessions, players started looking for me. Where was Dave? Was he setting up cones for the conditioning component of the sessions that typically followed the session? Smiles ensued, as they saw no cones out. "Yes, today we have no running," you could hear being whispered among the group.

"OK, cool down," I shouted. The girls started the two-lap cooldown that preempted team stretching. We wanted the athletes to feel relaxed, happy, safe, and comfortable in their environment. We wanted them to believe this was the end of training. Only three or four athletes had made it back to the bench before the head coach blew the whistle, sending all the players to the goal line, where I was waiting. The staff rapidly set up the cones. We had caught the team off guard, as planned. Questions ensued from the player group, demanding to know what was up. As agreed by the coaching staff, no information was to be given to the team pertaining to session details, except for the prescription of the run format, which was 10:10 at 102% straight, with a straight repetition followed by a shuttle, repeat until we say stop.

The athletes set themselves up into their respective 30-15 IFT groups for the conditioning (level 19.5, 20, 20.5, etc.) so the correct running distances were assigned. "How many are we doing?" a couple of athletes demanded.

Ignoring all questions, I simply shouted, "Ready?" The whistle went and the dislocation of expectation session began. I could still hear the players talking among themselves. "He normally does 12 reps of this, so worst case 14," was one comment.

The group was well conditioned and at this stage in our Olympic prep, the average 30-15 IFT score for the women's Olympic team was 20.8. This was an aerobically gifted group of athletes and I was unsure how many repetitions and sets of 10:10 at 102% it was going to take. It was the one unknown about this session experience that we couldn't account for. I suggested 15 to 17 repetitions would be uncomfortable, and that we would start to see substantial lagging in 20% to 30% of the group by then. I could not have been more wrong, and we as a staff could not have hoped for a better response and showing of resolve by the team.

After 22 repetitions, athletes started to fall to their knees during the rest periods, with players trying to motivate one another, yet barely getting more than one or two syllables out. Despite that, I counted down from 5. The athletes dragged themselves and each other off their knees to their feet for number 23. Their resolve and intent to not give up was impressive, but that was enough for that series. I faked the whistle and shouted, "Stop." The players collapsed to the ground.

That is it, it is over, they must have thought. "Three minutes rest," I shouted.

The staff was looking at each other and the players to see their response. Within 2 min, they were back to their feet and ready to go again. The mental strength and physical resilience to put themselves back on the line was the sole purpose of the session, and not the associated type 4 stimulus it had. That ability to go beyond what would be considered normal (12 to 14 repetitions in this case) and into a much darker place, a place they didn't think possible (22 repetitions), was the real aim. Perhaps this is such a place an athlete might find herself in during an Olympic match with a medal on the line. How much resolve can be found in the last 2 or 3 min, when you've already given everything and feel exhausted? What do you really have left to give for your team? Do you really know if you have it in you? As staff, we now knew that they really did.

I blew the whistle and we started again. But we had seen what we wanted. I stopped the session at 8 repetitions, and the players crumbled to the floor a second time. This time nothing was actually next. Just home on the bus, but with a new sense of what was now possible, and with a greater understanding of what hard really is.

The team medaled at the Olympics, achieving the first field hockey team medal in 20 years.